

WYSIMWYG: Simulation-based Decision Support for Data Distribution of Hyperlocal Services in Edge Cloud Environments

Sujeong Lim*
School of Computing, KAIST
Daejeon, Republic of Korea
sujeong@kaist.ac.kr

KyeongDeok Baek
School of Computing, KAIST
Daejeon, Republic of Korea
kyeongdeok.baek@kaist.ac.kr

In-Young Ko
School of Computing, KAIST
Daejeon, Republic of Korea
iko@kaist.ac.kr

Abstract

This study addresses effective decision-making for developers managing distributed storage resources in edge-cloud environments, particularly balancing data distribution for hyperlocal services. These neighborhood-focused services can leverage the edge-cloud environment to enhance performance and efficiency. Through a formative study, we identified key challenges, including developers' difficulty specifying performance and cost requirements due to uncertainty about outcomes and the need for human involvement alongside programmatic processes. Consequently, we developed WYSIMWYG, a simulation-based decision-support tool for data distribution in hyperlocal services operating within edge-cloud environments. WYSIMWYG provides simulation results to address uncertainties, map-based visualizations, and region-specific strategy customizations. A user study with 16 practitioners demonstrated that WYSIMWYG significantly improves decision confidence and facilitates intuitive decision-making.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Empirical studies in HCI*; • **Software and its engineering** → *Cloud computing*.

Keywords

Decision Support, Simulation, Hyperlocal Service, Edge Computing, Data Distribution

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1 Introduction

Hyperlocal services [12, 17], which cater to users' immediate communities, are gaining prominence by providing localized information and neighborhood interactions [19]. Examples include online

secondhand marketplaces such as Nextdoor¹ and Karrot², which enable users to trade goods within their local areas.

As these services grow, efficiently managing storage and computing resources becomes increasingly challenging. The edge-cloud computing paradigm, which integrates local storage and computing resources with centralized cloud infrastructure, has emerged as a promising solution [21]. It offers advantages such as reduced communication latency, alleviated network bottlenecks, and enhanced resource distribution to local edge nodes, enabling scalable and efficient resource management [27].

Given the geographical nature of hyperlocal services, where “data locality” links data to users' regions [10], leveraging edge-cloud environments can improve resource efficiency and enhance service performance by allowing users to access data from nearby storage locations [8, 25], as depicted in Figure 1.

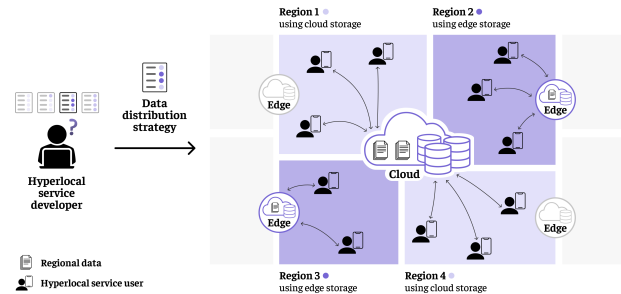


Figure 1: A developer's decision-making of the data distribution strategy for a hyperlocal service in an edge-cloud environment

However, existing research on edge-cloud environments primarily focuses on deterministic methods for resource allocation [24, 25]. These methods often fail to account for hyperlocal applications' unique, service-specific requirements. To address these limitations, we conducted a formative study to explore developers' perspectives, practical challenges, and key considerations. The findings highlight the critical role of human involvement in decision-making, the key considerations of cost and performance influenced by the balance between edge and cloud resources, and developers' difficulty specifying cost and performance requirements due to uncertainty about outcomes.

*Corresponding author

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¹<https://nextdoor.com/>

²<https://karrotmarket.com/>

Building on these insights, we propose WYSIMWYG, a simulation-based decision-support tool for data distribution in hyperlocal services operating within edge-cloud environments. WYSIMWYG enables developers to:

- Review simulation results for performance and cost of data distribution strategies before deployment.
- Customize strategies for specific regions based on service requirements.
- Visualize data distribution strategies through a map-based interface.

A user study involving 16 participants demonstrated that WYSIMWYG effectively supports decision-making, increasing confidence and usability compared to conventional cloud services.

The main contributions of this study are as follows:

- We identify practical problems that developers face in managing edge-cloud resources for hyperlocal services through a formative study.
- We develop WYSIMWYG, a simulation-based decision support tool to address these challenges, which remains underexplored in edge-cloud environments.
- We demonstrate WYSIMWYG's effectiveness in enhancing decision confidence and intuitive decision-making through a user study.

2 Related Work

2.1 Resource Allocation and Management in Cloud Environments

Resource management in cloud and edge environments has been widely studied, addressing resource allocation [2, 14, 22, 23, 28] and data caching [16, 24]. Most research employs static, optimization-based approaches, often focusing on minimizing costs through mathematical models and heuristics. While these methods are effective for programmatically determining which data to store on edge servers, they struggle to accommodate diverse service-specific requirements. For instance, a startup may prioritize cost efficiency, while a large enterprise may focus on high service quality. Therefore, this work proposes a decision-support tool that integrates human involvement to account for diverse service contexts.

2.2 Simulation and Forecasting for Decision Making

Simulation and forecasting methods have been proven effective for supporting decision-making by modeling potential outcomes. For example, Guo et al. [11] demonstrated that providing information about alternatives significantly boosts decision-makers' confidence. Similarly, AHA! [5] showed that simulating the potential harms of AI systems helps developers anticipate and mitigate risks. In this study, we apply simulation methods to data distribution in edge-cloud environments investigating their impact on and benefits for developers' decision-making processes.

3 Formative Study

To explore developers' considerations and challenges in using edge-cloud environments for hyperlocal services, we conducted a formative study involving 16 participants (4 males, 12 females, mean age

= 26.9), including 11 developers with cloud platform experience. The study aimed to assess the necessity of human involvement in decision-making and identify practical challenges from practitioners' perspectives. The survey questions of the formative study are described in the Appendix Table 1.

3.1 Results

3.1.1 Human involvement is essential. Participants agreed that data distribution requirements depend on service scale, financial constraints, and organizational policies. Automated approaches alone are insufficient to address these diverse and service-specific requirements due to the unique nature of each service. These findings underscore the importance of involving developers in decision-making regarding data distribution, rather than relying solely on programmatic processes.

3.1.2 The primary considerations in this decision-making process are cost and performance. Participants supported this with their experience in cloud resource management, noting that such decisions generally involve evaluating cost-effectiveness and comparing the performance specifications of various cloud platforms.

3.1.3 Decision-makers often encounter difficulties in specifying desired cost and performance requirements. When asked about these difficulties, participants noted challenges in determining the appropriate performance level for specific services and predicting the actual costs suitable for the service context, primarily due to uncertainties regarding the outcomes of data distribution.

This paper focuses on the challenges faced by developers of hyperlocal services in utilizing edge-cloud environments, particularly the complexities involved in selecting an optimal data distribution strategy. We aim to propose a decision support tool for data distribution in edge-cloud environments, using simulation methodology to address these challenges.

4 WYSIMWYG: Simulation-based Decision Support for Data Distribution

Building on insights from our formative study, we developed WYSIMWYG. This simulation-based decision support tool helps developers identify appropriate data distribution strategies for hyperlocal services in edge-cloud environments. "WYSIMWYG" stands for "What You Simulate is What You Get," emphasizing that developers can simulate data distribution strategies and observe the anticipated outcomes before the actual distribution.

As illustrated in Figure 2a, WYSIMWYG incorporates four key features.

- A. Level-Based Control Interface:** Allows developers to control the ratio of edge storage usage.
- B. Regional Strategy Customization:** Enables fine-tuning data distribution strategies for specific regions.
- C. Map-Based Visualization of Data Distribution:** Provides a visual representation of the applied data distribution strategy across service regions.
- D. Simulation Results Provision:** Offers performance and cost estimates for the selected data distribution strategy.

The tool's user interface is designed for a hypothetical hyperlocal service, ensuring practical relevance.

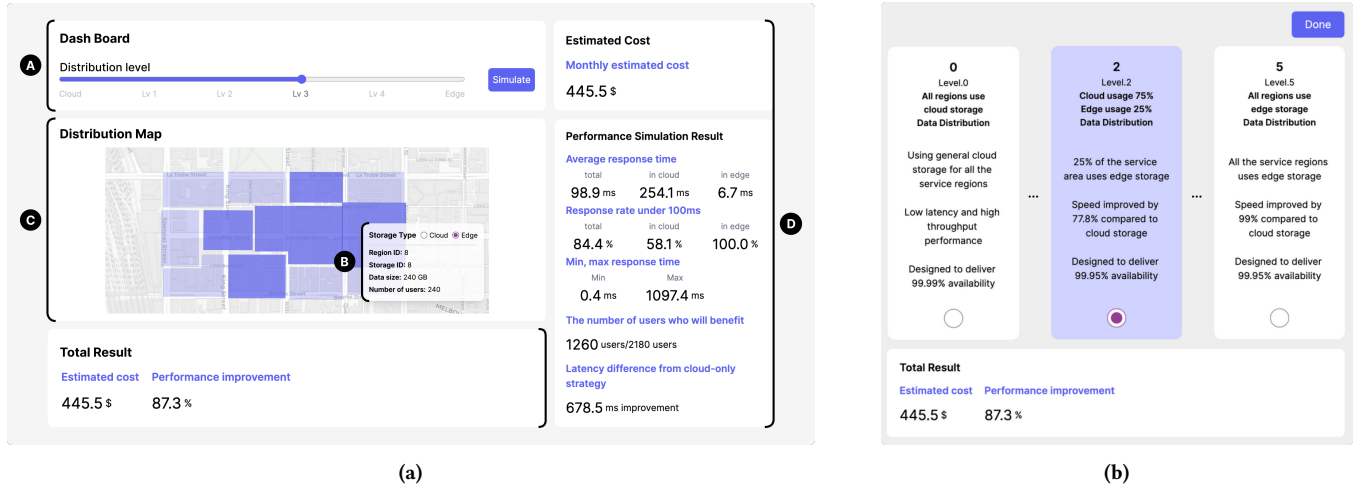


Figure 2: (a) User interface of WYSIMWYG. (b) User interface of the baseline approach.

4.1 Level-based Control

The level-based control feature simplifies decision-making by allowing developers to determine the global ratio of edge storage utilization. This functionality reduces cognitive load by eliminating the need to input specific numerical values.

As shown in Figure 2a-A, WYSIMWYG offers six levels, ranging from the *Cloud* level (far left), which represents the exclusive use of cloud storage, indicating traditional cloud environments, to the *Edge* level (far right), which represents the exclusive use of edge storage across all regions. Higher levels indicate increased edge storage utilization, aligning with more distributed strategies. Developers can explore these levels to select the most suitable strategy.

4.2 Regional Strategy Customization

In addition to the level-based control, developers can customize strategies for individual regions. Given the region-specific nature of hyperlocal services, the service area is segmented into smaller regions. Developers can use the interface (Figure 2a-B) to select each region's storage resource type—cloud or edge. This customization feature expands options beyond coarse, global control, allowing developers to reflect region-specific preferences and policies for data distribution.

4.3 Map-based Visualization of Data Distribution

The presentation of information heavily influences effective decision making [3, 9, 26]. Dennis et al. [6] have shown that map-based presentations are more effective in decision-making of geographic task than tabular formats. Moreover, Hwang et al. [13] demonstrated that spatial presentations lead to more accurate decision-making in complex tasks. WYSIMWYG employs a map-based interface (Figure 2a-C) to visualize data distribution strategies across regions. Each region is depicted as a colored box, with lighter shades representing cloud storage usage and bolder shades indicating edge storage usage. This intuitive visualization enables developers to

comprehend and refine their strategies more effectively, making the process more engaging.

4.4 Provision of Simulation Results

Decision support research underscores the importance of providing alternatives to improve decision-making [5, 7, 11]. WYSIMWYG fulfills this requirement by allowing developers to simulate and evaluate different data distribution strategies. Based on the strategy chosen through the above features, WYSIMWYG estimates performance and cost outcomes, as shown in Figure 2a-D. The simulation results presented in WYSIMWYG focus on performance and cost, which our formative study identified as important considerations. Key metrics include: (1) average response time, (2) response rate under 100ms [1], (3) minimum and maximum response times, (4) the number of users benefiting from the distribution strategy, (5) latency differences compared to a cloud-only strategy, (6) total improvement rate, and (7) estimated cost. The storage costs used to obtain the estimated cost are based on the pricing policy of the existing cloud platform service³. Performance simulation results are derived from the outcomes of the simulator, which we implemented using Yet Another Fog Simulator (YAFS), one of the most popular simulators for edge-cloud computing [15].

The source code of WYSIMWYG is available in the public repository⁴. WYSIMWYG and simulation results presented in this paper are tailored to a particular hyperlocal service designed for the user study. However, WYSIMWYG can be adapted for any hyperlocal service, provided that service providers supply the necessary information of their services.

5 Evaluation

We conducted a user study with practitioners to assess WYSIMWYG's effectiveness and usability and the impact of simulation on addressing our target problem. Our university approved the study through the Institutional Review Board. This section details the

³<https://aws.amazon.com/s3/pricing/>

⁴<https://github.com/sio2whocodes/WYSIMWYG>

experiment design, procedures, and analysis of the results, including participants' feedback.

5.1 Participants

We used snowball sampling to recruit 16 participants (10 females, 6 males, mean age = 27) with experience using cloud platforms such as Amazon Web Services. Of these, 11 participants are developers with an average of 1.9 years of work experience, while five are graduate students majoring in computer science. Notably, three participants have experience in developing hyperlocal services. Each participant received approximately USD 15 for one hour of participation.

5.2 User Tasks

The task for participants was to determine an appropriate data distribution strategy for a hypothetical hyperlocal service in an edge-cloud environment. Specifically, as shown in Figure 1, this involved deciding how to balance the use of edge and cloud resources. Participants were asked to imagine themselves as follows: *You are a developer working on a hyperlocal service, and try to apply edge-cloud environments in your service to distribute regional data. You need to determine an efficient data distribution strategy.* Three distinct scenarios were presented, each varying in service scale, cost constraints, and performance goals:

- Scenario A: Developing a hyperlocal service for the company participants currently or previously worked for, allowing for an immersive context.
- Scenario B: Developing a hyperlocal service in a large corporation, where the primary goal of using edge-cloud environments is to improve service quality, supported by sufficient financial resources.
- Scenario C: Developing a hyperlocal service for a startup with limited financial resources.

5.3 Experiment Design and Procedure

The user study employed a within-subject design, while the decision support approach (WYSIMWYG vs. baseline) was the independent variable. The baseline is implemented to resemble a resource management interface in conventional cloud environments. As shown in Figure 2b, it presents information on six distribution strategies, based on the ratio of edge resource usage, in a text-based list format.

Participants first completed a pre-study survey to provide demographic information, details about their development experience, and familiarity with cloud platform services. We explained the research background, including the concept of edge-cloud environments, data distribution in these environments, and hyperlocal services, supplemented with examples. After that, participants were asked to make a decision on the strategy regarding the appropriate data distribution for each of the three service scenarios using WYSIMWYG and the baseline approach. The order of using two approaches is counterbalanced. After the decision-making using each tool, participants completed a post-study survey evaluating their experience with the tool. This survey included (1) SUS questionnaire [4], (2) USE questionnaire [18], and (3) our custom questions asking about their experience with the tool on deciding the proper

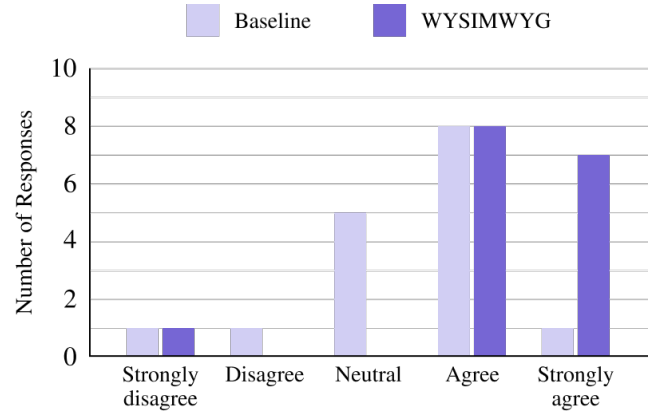


Figure 3: Response distribution to the question on decision Confidence, ‘Do you believe you can get the desired result with the decision made using the tool?’

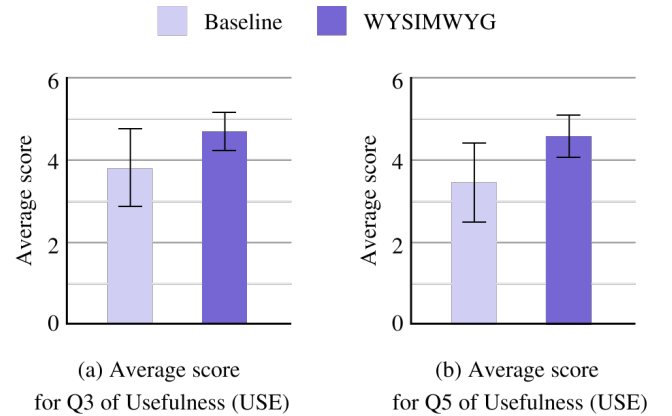


Figure 4: Responses to usefulness questions (Q3 and Q5) from the USE questionnaire

strategy. Our custom questions are described in the Appendix Table 2. Finally, we concluded the user study with the final survey consisting of detailed questions about their experience with WYSIMWYG and its components. The questions examined how each feature influences decision-making.

5.4 Results

The user study evaluated WYSIMWYG’s effectiveness in supporting data distribution strategy decisions in edge-cloud environments for hyperlocal services. The evaluation criteria were confidence, usefulness, decision changes, and usability.

5.4.1 Confidence. To evaluate confidence in decision-making, participants were asked whether they believed they could achieve their desired results for data distribution using each tool. The result reveals a significant difference ($p < 0.05$) in participants’ confidence levels, with more dispersed responses for the baseline approach, as shown in Figure 3. This result indicated that WYSIMWYG enables participants to make decisions with greater confidence. As P10

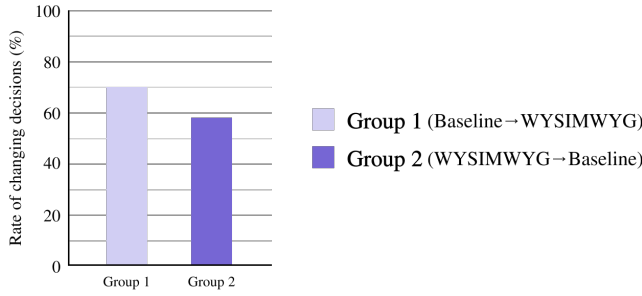


Figure 5: Decision change rates

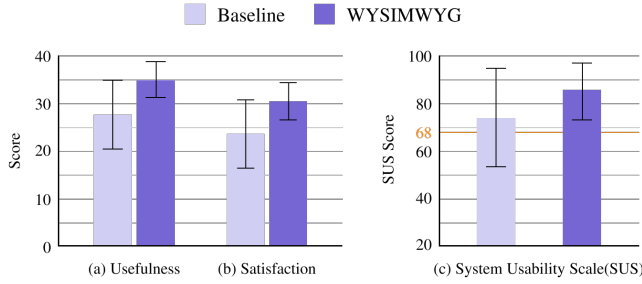


Figure 6: Results of usability evaluation

remarked about the baseline method, “I feel it is hard to expect to get the precise result because it (the baseline approach) provides less information. I made my choice with the possibility that the predictions might not be accurate.” This feedback underscores how the limited information provided by conventional methods in a cloud environment can undermine users’ confidence in decision-making for edge-cloud scenarios.

5.4.2 Usefulness. The usefulness of each approach was assessed through responses to two questions from the USE questionnaire [18]:

Q3. It is useful.

Q5. It makes the things I want to accomplish easier to get done.

The results in Figures 4 (a) and 4 (b) indicate that participants found WYSIMWYG more useful than the baseline approach. For Q3, WYSIMWYG received a higher mean score ($mean = 4.6, SD = 0.46$) compared to the baseline ($mean = 3.8, SD = 0.95$), with a statistically significant difference ($p < .005$). Similarly, for Q5, WYSIMWYG was rated higher ($mean = 4.6, SD = 0.5$) than the baseline ($mean = 3.4, SD = 0.93$), showing a significant difference ($p < .001$). These findings indicated that WYSIMWYG is more effective in facilitating decision-making. Participants provided qualitative feedback supporting these results. P5 stated, “By comparing various values through detailed simulation results, I can make a decision more easily.” P4 noted, “I could estimate the extent of improvement by looking at the map and found the visualization made it easier to predict.”

5.4.3 Decision changes. To further assess WYSIMWYG’s impact on decision-making, we analyzed the rate of decision changes based on the sequence of tool usage (WYSIMWYG and baseline). Although the user study followed a within-subject design, this analysis divided

participants into two groups for a between-subject comparison: Group 1 used the baseline first, followed by WYSIMWYG and Group 2 used these tools in reverse order. The analysis of decision change rates, illustrated in Figure 5, revealed that Group 2 was less likely to change the decision (58%) than Group 1 (70%). This indicated that WYSIMWYG enables participants to recognize discrepancies between the strategies chosen using the baseline, aligning more closely with their desired outcomes, as supported by concrete simulation results. For example, when using WYSIMWYG for scenario B, P4 remarked, “I will never choose level 2 (that P4 chose when using the baseline) if the performance result would be like this.” This comment illustrates how performance simulation results provided by WYSIMWYG guide users toward more appropriate decisions. Ultimately, WYSIMWYG aligns participants’ decisions more closely with their initial intentions, emphasizing its value in supporting effective decision-making.

5.4.4 Usability. The usability of WYSIMWYG was evaluated using the SUS and USE questionnaires. Figure 6 (c) illustrates that WYSIMWYG demonstrates substantially better usability than the baseline, achieving an above-average usability score [20] on the SUS questionnaire. Results from the usefulness and satisfaction sections of the USE questionnaire are represented in Figures 6 (a) and 6 (b), respectively. The findings indicated that WYSIMWYG exhibits significantly higher usability, with statistical significance in perceived usefulness ($F = 18.14, p < 0.001$) and satisfaction ($F = 14.33, p < 0.005$). Overall, WYSIMWYG outperforms the baseline approach in usability, even though the baseline has fewer steps in the decision-making process. P11 remarked, “The tool (WYSIMWYG) is straightforward and easy to use. It seems that even first-time users will be able to use it easily.” Furthermore, P10, P11, and P13 highlighted that the region presentation in WYSIMWYG was intuitive and visually clear, enhancing its overall usability.

5.4.5 Impact of WYSIMWYG’s Features. In addition to evaluating the effectiveness of WYSIMWYG, we also investigated the impact of each feature by asking participants to select features they would be most likely to use in a real-world scenario from the following three main features: (1) level-based control, (2) regional strategy customization, and (3) provision of simulation results. The most preferred feature, chosen by all participants, is the provision of simulation results, followed by regional strategy customization with 13 choices as the second most preferred. As the results demonstrate, providing simulation results was crucial in enhancing WYSIMWYG’s effectiveness.

6 Conclusion

This study focuses on developers’ decision-making regarding data distribution strategies for hyperlocal services in edge-cloud environments. To understand practitioners’ perspectives, we conducted a formative study that provides practical insights into the necessity of human involvement, key considerations, and challenges in decision-making. We developed WYSIMWYG, a simulation-based decision support tool for data distribution of hyperlocal services in edge-cloud environments. WYSIMWYG addresses decision-making uncertainties by providing simulation results for various data distribution strategies, an approach that remains underexplored in this domain. Additionally, WYSIMWYG incorporates decision support

techniques such as map-based visualization, level-based control, and region-specific customization tailored to hyperlocal services.

A user study was conducted to evaluate the effectiveness of WY-SIMWYG in supporting decision-making for appropriate data distribution strategies. The user study revealed that WY-SIMWYG enhances decision-making by substantially improving decision confidence and perceived usefulness compared to the baseline. Participants highlighted that the tool's concrete simulation results bolstered their confidence, while its map-based visualization enabled them to intuitively understand data distribution strategies. In conclusion, WY-SIMWYG offers robust features that empower hyperlocal service developers to efficiently manage storage resources in edge-cloud environments. Our empirical findings would contribute to researchers investigating the interactions necessary for resource management in edge-cloud environments. For future work, we plan to conduct multiple design iterations based on user feedback, focusing on the addition of strategy recommendations and a more direct strategy comparison feature.

Acknowledgments

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References

- [1] Christiane Attig, Nadine Rauh, Thomas Franke, and Josef F. Krems. 2017. System Latency Guidelines Then and Now – Is Zero Latency Really Considered Necessary?. In *Engineering Psychology and Cognitive Ergonomics: Cognition and Design* (Cham), Don Harris (Ed.). Springer International Publishing, Cham, 3–14.
- [2] Ali Belgacem, Kadda Beghdad-Bey, Hassina Nacer, and Sofiane Bouznad. 2020. Efficient dynamic resource allocation method for cloud computing environment. *Cluster Computing* 23, 4 (2020), 2871–2889.
- [3] Izak Benbasat, Albert S Dexter, and Peter Todd. 1986. An experimental program investigating color-enhanced and graphical information presentation: an integration of the findings. *Commun. ACM* 29, 11 (1986), 1094–1105.
- [4] John Brooke. 1995. SUS: A quick and dirty usability scale. *Usability Eval. Ind.* 189 (1995).
- [5] Zana Bućinca, Chau Minh Pham, Maurice Jakesch, Marco Tulio Ribeiro, Alexandra Olteanu, and Saleema Amershi. 2023. AHA!: Facilitating AI Impact Assessment by Generating Examples of Harms. arXiv:2306.03280 [cs.HC] <https://arxiv.org/abs/2306.03280>
- [6] Alan R Dennis and Traci A Carte. 1998. Using geographical information systems for decision making: Extending cognitive fit theory to map-based presentations. *Information systems research* 9, 2 (1998), 194–203.
- [7] Evanthis Dimara and John Stasko. 2022. A Critical Reflection on Visualization Research: Where Do Decision Making Tasks Hide? *IEEE Transactions on Visualization and Computer Graphics* 28, 1 (2022), 1128–1138. <https://doi.org/10.1109/TVCG.2021.3114813>
- [8] Utsav Drolia, Rolando Martins, Jiaqi Tan, Ankit Chheda, Monil Sanghavi, Rajeev Gandhi, and Priya Narasimhan. 2013. The Case for Mobile Edge-Clouds. In *2013 IEEE 10th International Conference on Ubiquitous Intelligence and Computing and 2013 IEEE 10th International Conference on Autonomic and Trusted Computing*. IEEE, Vietri sul Mare, Italy, 209–215. <https://doi.org/10.1109/UIC-ATC.2013.94>
- [9] Karin Eberhard. 2023. The effects of visualization on judgment and decision-making: a systematic literature review. *Management Review Quarterly* 73, 1 (2023), 167–214.
- [10] Tadeu Freitas, João Rodrigues, Diogo Bogas, Miguel Coimbra, and Rolando Martins. 2018. Panoptic, Privacy over Edge-Clouds. In *2018 IEEE 6th International Conference on Future Internet of Things and Cloud (FiCloud)*. IEEE, Barcelona, 325–332. <https://doi.org/10.1109/FiCloud.2018.00054>
- [11] Shunan Guo, Fan Du, Sana Malik, Eunyee Koh, Sungchul Kim, Zhicheng Liu, Donghyun Kim, Hongyuan Zha, and Nan Cao. 2019. Visualizing Uncertainty and Alternatives in Event Sequence Predictions. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300803>
- [12] Nadav Hochman, Lev Manovich, and Mehrdad Yazdani. 2014. On hyper-locality: performances of place in social media. In *Proceedings of 2014 International AAAI Conference on Weblogs and Social Media*. AAAI Press, Ann Arbor, Michigan USA.
- [13] Mark IH Hwang and Bruce JP Wu. 1991. The effectiveness of computer graphics for decision support: meta-analytical integration of research findings. *ACM SIGMIS Database: The DATABASE for Advances in Information Systems* 21, 2-3 (1991), 11–20.
- [14] Weiwei Kong, Yang Lei, and Jing Ma. 2016. Virtual machine resource scheduling algorithm for cloud computing based on auction mechanism. *Optik* 127, 12 (2016), 5099–5104.
- [15] Isaac Lera, Carlos Guerrero, and Carlos Juiz. 2019. YAFS: A simulator for IoT scenarios in fog computing. *IEEE Access* 7 (2019), 91745–91758.
- [16] Ying Liu, Qiang He, Dequan Zheng, Xiaoyu Xia, Feifei Chen, and Bin Zhang. 2022. Data Caching Optimization in the Edge Computing Environment. *IEEE Transactions on Services Computing* 15, 4 (2022), 2074–2085. <https://doi.org/10.1109/TSC.2020.3032724>
- [17] Claudia López and Rosta Farzan. 2015. From Community Networks to Hyper-local Social Media. In *Proceedings of the 18th ACM Conference Companion on Computer Supported Cooperative Work & Social Computing* (Vancouver, BC, Canada) (CSCW'15 Companion). Association for Computing Machinery, New York, NY, USA, 239–242. <https://doi.org/10.1145/2685553.2699016>
- [18] Arnold Lund. 2001. Measuring Usability with the USE Questionnaire. *Usability and User Experience Newsletter of the STC Usability SIG* 8 (01 2001).
- [19] William P. Rey, Rayven Joseph V. Mercado, James Andrei C. Rivera, and Kenneth AJ G. Quilantang. 2022. E-ULAT: A web-based Community-Driven Application for Hyperlocal Community. In *Proceedings of the 4th International Conference on Management Science and Industrial Engineering* (Chiang Mai, Thailand) (MSIE '22). Association for Computing Machinery, New York, NY, USA, 414–420. <https://doi.org/10.1145/3535782.3535836>
- [20] Jeff Sauro. 2011. Measuring usability with the system usability scale. <https://measuringu.com/sus/>
- [21] Weisong Shi, Jie Cao, Quan Zhang, Youhuizi Li, and Lanyu Xu. 2016. Edge Computing: Vision and Challenges. *IEEE Internet of Things Journal* 3, 5 (2016), 637–646. <https://doi.org/10.1109/JIOT.2016.2579198>
- [22] Jinn-Tsong Tsai, Jia-Cen Fang, and Jyh-Horng Chou. 2013. Optimized task scheduling and resource allocation on cloud computing environment using improved differential evolution algorithm. *Computers & Operations Research* 40, 12 (2013), 3045–3055.
- [23] Wei-Jen Wang, Yue-Shan Chang, Win-Tsung Lo, and Yi-Kang Lee. 2013. Adaptive scheduling for parallel tasks with QoS satisfaction for hybrid cloud environments. *The Journal of Supercomputing* 66 (2013), 783–811.
- [24] Xiaoyu Xia, Feifei Chen, Qiang He, John Grundy, Mohamed Abdelrazek, and Hai Jin. 2021. Online Collaborative Data Caching in Edge Computing. *IEEE Transactions on Parallel and Distributed Systems* 32, 2 (2021), 281–294. <https://doi.org/10.1109/TPDS.2020.3016344>
- [25] Xiaoyu Xia, Feifei Chen, Qiang He, John C Grundy, Mohamed Abdelrazek, and Hai Jin. 2020. Cost-effective app data distribution in edge computing. *IEEE Transactions on Parallel and Distributed Systems* 32, 1 (2020), 31–44.
- [26] Qian Yang, Richmond Y. Wong, Thomas Gilbert, Margaret D. Hagan, Steven Jackson, Sabine Junginger, and John Zimmerman. 2023. Designing Technology and Policy Simultaneously: Towards A Research Agenda and New Practice. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 343, 6 pages. <https://doi.org/10.1145/3544549.3573827>
- [27] Shanhe Yi, Cheng Li, and Qun Li. 2015. A Survey of Fog Computing: Concepts, Applications and Issues. In *Proceedings of the 2015 Workshop on Mobile Big Data* (Hangzhou, China) (Mobidata '15). Association for Computing Machinery, New York, NY, USA, 37–42. <https://doi.org/10.1145/2757384.2757397>
- [28] Liyun Zuo, Lei Shu, Shoubin Dong, Chunsheng Zhu, and Takahiro Hara. 2015. A multi-objective optimization scheduling method based on the ant colony algorithm in cloud computing. *Ieee Access* 3 (2015), 2687–2699.

A Survey Questions

Table 1: The survey questions of the formative study

	Questions
Q1	When considering a tool for managing data distribution in edge-cloud environments, do you believe that the tool should enable developers or service providers to select computational resource capacity based on service-specific factors, such as service scale and financial constraints?
Q2	What would be the most important factors you would consider when choosing a strategy? (options: cost, stability performance, speed performance, security, or other factors)
Q3	Do you think it is easy to specify the desired cost for data distribution?
Q4	Do you think it is easy to specify the desired service performance achieved through data distribution?

Table 2: Our custom questions of the post-study survey in the user study

	Questions
Q1	I believe that the chosen distribution strategy using the tool will produce the intended outcomes for each scenario.
Q2	Please freely describe the reason for your answer to Q1.
Q3	It was easy to determine a data distribution strategy suitable for the hypothetical scenario using this tool.
Q4	I would like to use this tool if it were available in real-world situations.